Unit Test Report

Student Attendance Management System (SAMS)

Project	Student Attendance Management System (SAMS)
Version	1.0
Authors	Atul Kandiyil (PES2UG24CS920), Riona Jeena Dsouza (PES2UG24CS654), Amit (PES2UG24CS624) - Team 1
Date	06-11-2026
Build tested	main branch, commit 33cdb28, plus the new test files
Environment	Python 3.12.3, Django 5.2.18, SQLite, Linux

Revision History

Version	Date	Author	Change
1.0	06-11-2026	Team 1	First version.

Approvals

Role	Name	Signature / Date
Team Lead	Atul Kandiyil	
Dev Lead	Riona Jeena Dsouza	
Test Engineer	Amit	

1. Introduction

This report covers the automated unit and integration tests for SAMS. They are run with Django's test runner (`python manage.py test`) and were run in the testing week as required by the Test Plan. The test cases they belong to are in the Test Case Document.

2. Summary

Tests run	83
Passed	81
Expected failures	2
Unexpected failures or errors	0
Tests from Sprint 1 and 2	20
Tests added in this round	63
Run time	about 1.6 seconds

All 83 tests ran. 81 passed and nothing failed unexpectedly. The 2 expected failures are tests we wrote for two bugs found during this round (DEF-01 and DEF-02, described in the System Validation Report). We marked them with `@unittest.expectedFailure` so the GitHub Actions run stays green while the bugs are open. After the bugs are fixed these two will show as "unexpected success" and the decorator can be removed.

3. Results by module

Module	Tests	Passed	Expected failures	Failed
accounts	21	20	1	0
administration	12	12	0	0
attendance	50	49	1	0

4. Results by test class

Test class	Tests	Passed	Expected failures	Failed
accounts.tests_auth.DashboardTests	6	6	0	0
accounts.tests_auth.LoginLogoutTests	7	7	0	0
accounts.tests_auth.PasswordAndTransportSecurityTests	3	2	1	0
accounts.tests_auth.RoleBasedAccessTests	5	5	0	0
administration.tests.EnrolmentConstraintTest	1	1	0	0
administration.tests.StudentModelTest	1	1	0	0
administration.tests_admin.AdminManagementTests	10	10	0	0
attendance.tests.AttendanceModelTest	2	2	0	0
attendance.tests.AttendanceWorkflowTests	3	3	0	0
attendance.tests.Sprint2Tests	13	13	0	0
attendance.tests_regression.HealthCheckFailureTests	1	1	0	0
attendance.tests_regression.MarkingTests	11	10	1	0

attendance.tests_regression.ModelAndServiceTests	2	2	0	0
attendance.tests_regression.ReportViewTests	13	13	0	0
attendance.tests_regression.SessionCreationTests	5	5	0	0

5. Tests that are expected to fail

Defect	Test	What happens
DEF-01	attendance.tests_regression.MarkingTests.test_tc_data_01_partial_submission_is_rolled_back	One record is left in the database when 0 are expected
DEF-02	accounts.tests_auth.PasswordAndTransportSecurityTests.test_tc_sec_04_secure_cookies_enabled	SESSION_COOKIE_SECURE is False because DEBUG is hard-coded to True

6. All tests

No.	Test	Result	Time (s)
1	accounts.tests_auth.DashboardTests.test_admin_dashboard_shows_totals	Pass	0.028
2	accounts.tests_auth.DashboardTests.test_faculty_dashboards_show_assigned_subjects_only	Pass	0.014
3	accounts.tests_auth.DashboardTests.test_student_dashboard_with_profile_and_records	Pass	0.009
4	accounts.tests_auth.DashboardTests.test_student_dashboard_without_attendance_records	Pass	0.009
5	accounts.tests_auth.DashboardTests.test_student_dashboard_without_student_profile	Pass	0.007
6	accounts.tests_auth.DashboardTests.test_user_str	Pass	0.002
7	accounts.tests_auth.LoginLogoutTests.test_tc_auth_01_login_page_renders	Pass	0.003
8	accounts.tests_auth.LoginLogoutTests.test_tc_auth_01_role_redirect_per_role	Pass	0.021
9	accounts.tests_auth.LoginLogoutTests.test_tc_auth_01_valid_login_creates_session	Pass	0.004
10	accounts.tests_auth.LoginLogoutTests.test_tc_auth_02_error_message_is_generic	Pass	0.005
11	accounts.tests_auth.LoginLogoutTests.test_tc_auth_02_invalid_password_rejected	Pass	0.003
12	accounts.tests_auth.LoginLogoutTests.test_tc_auth_02_unknown_user_rejected	Pass	0.004
13	accounts.tests_auth.LoginLogoutTests.test_tc_auth_04_logout_ends_session	Pass	0.011
14	accounts.tests_auth.PasswordAndTransportSecurityTests.test_tc_sec_03_password_is_hashed_not_plaintext	Pass	0.761
15	accounts.tests_auth.PasswordAndTransportSecurityTests.test_tc_sec_04_proxy_ssl_header_configured	Pass	0.001
16	accounts.tests_auth.PasswordAndTransportSecurityTests.test_tc_sec_04_secure_cookies_enabled	Expected fail	0.002
17	accounts.tests_auth.RoleBasedAccessTests.test_tc_auth_03_admin_allowed_on_admin_pages	Pass	0.015
18	accounts.tests_auth.RoleBasedAccessTests.test_tc_auth_03_faculty_blocked_from_student_and_admin_pages	Pass	0.008
19	accounts.tests_auth.RoleBasedAccessTests.test_tc_auth_03_student_blocked_from_faculty_and_admin_pages	Pass	0.011
20	accounts.tests_auth.RoleBasedAccessTests.test_tc_sec_01_unauthenticated_redirected_to_login	Pass	0.005
21	accounts.tests_auth.RoleBasedAccessTests.test_tc_sec_02_student_cannot_post_attendance_directly	Pass	0.006
22	administration.tests.EnrolmentConstraintTest.test_duplicate_student_subject_enrolment_is_rejected	Pass	0.002
23	administration.tests.StudentModelTest.test_student_creation	Pass	0.001
24	administration.tests_admin.AdminManagementTests.test_audit_log_view_lists_events_for_admin	Pass	0.007
25	administration.tests_admin.AdminManagementTests.test_model_string_representations	Pass	0.005
26	administration.tests_admin.AdminManagementTests.test_tc_enr_01_enrolled_student_appears_in_marking_list	Pass	0.015
27	administration.tests_admin.AdminManagementTests.test_tc_fac_01_admin_creates_faculty_who_can_login	Pass	0.010
28	administration.tests_admin.AdminManagementTests.test_tc_fac_02_assigned_faculty_sees_subject	Pass	0.012
29	administration.tests_admin.AdminManagementTests.test_tc_stu_01_admin_adds_student	Pass	0.010
30	administration.tests_admin.AdminManagementTests.test_tc_stu_01_duplicate_student_id_rejected	Pass	0.006
31	administration.tests_admin.AdminManagementTests.test_tc_stu_02_admin_updates_student	Pass	0.011
32	administration.tests_admin.AdminManagementTests.test_tc_stu_03_admin_lists_students	Pass	0.026
33	administration.tests_admin.AdminManagementTests.test_tc_sub_01_subject_created_and_code_unique	Pass	0.004
34	attendance.tests.AttendanceModelTest.test_attendance_creation	Pass	0.002
35	attendance.tests.AttendanceModelTest.test_duplicate_attendance_is_rejected	Pass	0.002
36	attendance.tests.AttendanceWorkflowTests.test_faculty_can_open_attendance_page	Pass	0.008
37	attendance.tests.AttendanceWorkflowTests.test_faculty_can_save_attendance	Pass	0.013
38	attendance.tests.AttendanceWorkflowTests.test_student_cannot_mark_attendance	Pass	0.010
39	attendance.tests.Sprint2Tests.test_tc_att_03_modify_attendance_permitted	Pass	0.015
40	attendance.tests.Sprint2Tests.test_tc_att_03_modify_attendance_unauthorized_blocked	Pass	0.014
41	attendance.tests.Sprint2Tests.test_tc_audit_01_and_sec_05_audit_logging	Pass	0.021
42	attendance.tests.Sprint2Tests.test_tc_calc_01_attendance_percentage_calculation	Pass	0.004
43	attendance.tests.Sprint2Tests.test_tc_calc_02_low_attendance_identification	Pass	0.013

44	attendance.tests.Sprint2Tests.test_tc_hist_01_student_attendance_history	Pass	0.012
45	attendance.tests.Sprint2Tests.test_tc_not_01_low_attendance_notification	Pass	0.023
46	attendance.tests.Sprint2Tests.test_tc_perf_01_response_time	Pass	0.012
47	attendance.tests.Sprint2Tests.test_tc_rel_01_health_check_endpoint	Pass	0.003
48	attendance.tests.Sprint2Tests.test_tc_rep_01_attendance_report_criteria	Pass	0.017
49	attendance.tests.Sprint2Tests.test_tc_rep_02_attendance_summaries_and_csv_export	Pass	0.009
50	attendance.tests.Sprint2Tests.test_tc_view_01_student_view_own_attendance_only	Pass	0.016
51	attendance.tests.Sprint2Tests.test_tc_view_02_faculty_view_assigned_classes_only	Pass	0.012
52	attendance.tests_regression.HealthCheckFailureTests.test_tc_rel_01_health_reports_503_when_database_down	Pass	0.001
53	attendance.tests_regression.MarkingTests.test_invalid_status_rejected_with_400	Pass	0.010
54	attendance.tests_regression.MarkingTests.test_modification_without_reason_uses_default	Pass	0.015
55	attendance.tests_regression.MarkingTests.test_modify_view_shows_existing_values	Pass	0.014
56	attendance.tests_regression.MarkingTests.test_other_faculty_cannot_mark_or_modify	Pass	0.010
57	attendance.tests_regression.MarkingTests.test_tc_att_01_mark_all_statuses	Pass	0.012
58	attendance.tests_regression.MarkingTests.test_tc_att_03_modification_records_old_new_and_reason	Pass	0.016
59	attendance.tests_regression.MarkingTests.test_tc_att_04_resubmission_does_not_duplicate	Pass	0.014
60	attendance.tests_regression.MarkingTests.test_tc_data_01_database_blocks_duplicates	Pass	0.003
61	attendance.tests_regression.MarkingTests.test_tc_ux_01_bulk_marking_buttons_present	Pass	0.009
62	attendance.tests_regression.MarkingTests.test_unchanged_resubmission_writes_no_modification_audit	Pass	0.014
63	attendance.tests_regression.MarkingTests.test_tc_data_01_partial_submission_is_rolled_back	Expected fail	0.010
64	attendance.tests_regression.ModelAndServiceTests.test_attendance_and_session_str	Pass	0.003
65	attendance.tests_regression.ModelAndServiceTests.test_audit_log_anonymous_actor_and_strings	Pass	0.003
66	attendance.tests_regression.ReportViewTests.test_admin_sees_all_subjects_in_reports	Pass	0.011
67	attendance.tests_regression.ReportViewTests.test_defaulters_access_rules	Pass	0.020
68	attendance.tests_regression.ReportViewTests.test_defaulters_view_threshold_param	Pass	0.024
69	attendance.tests_regression.ReportViewTests.test_faculty_cannot_see_student_page	Pass	0.008
70	attendance.tests_regression.ReportViewTests.test_faculty_report_excludes_other_faculty_subjects	Pass	0.012
71	attendance.tests_regression.ReportViewTests.test_session_report_denied_for_other_faculty_and_student	Pass	0.019
72	attendance.tests_regression.ReportViewTests.test_session_report_for_faculty_and_admin	Pass	0.025
73	attendance.tests_regression.ReportViewTests.test_student_attendance_page_and_missing_profile	Pass	0.022
74	attendance.tests_regression.ReportViewTests.test_student_with_no_records_is_not_flagged	Pass	0.006
75	attendance.tests_regression.ReportViewTests.test_tc_calc_01_percentage_values	Pass	0.004
76	attendance.tests_regression.ReportViewTests.test_tc_calc_02_threshold_boundary	Pass	0.008
77	attendance.tests_regression.ReportViewTests.test_tc_rep_01_filters_subject_student_and_dates	Pass	0.031
78	attendance.tests_regression.ReportViewTests.test_tc_rep_02_summary_numbers_and_csv_match	Pass	0.012
79	attendance.tests_regression.SessionCreationTests.test_create_session_and_audit	Pass	0.010
80	attendance.tests_regression.SessionCreationTests.test_duplicate_session_not_created	Pass	0.008
81	attendance.tests_regression.SessionCreationTests.test_missing_date_rejected	Pass	0.009
82	attendance.tests_regression.SessionCreationTests.test_other_faculty_cannot_open_session_for_subject	Pass	0.008
83	attendance.tests_regression.SessionCreationTests.test_session_page_lists_sessions	Pass	0.008

7. Notes

While `manage.py test` runs, the password hasher is switched to MD5 to keep the run fast (Deviation 3). The password tests switch it back to PBKDF2, and the System Validation Report checks the hashes stored in the real database as well.

The raw output is saved next to this report as `unit_test_results.xml` (JUnit format) and `unit_test_log.txt`.